

SERVEID: Hierarchy-Aligned and Direction-Aware Knowledge Distillation for Semantic-ID Generative Recommendation

Supplementary Material

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1 Dataset Statistics

[TODO: fill in the Beauty/Toys/Video Games statistics (users, items, interactions, average sequence length, number of valid SIDs) in the same style as the previous submission.]

Dataset	#Users	#Items	#Interactions	Avg. len.
Amazon Beauty 2014	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Amazon Toys 2014	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Video Games 2023	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]

Table 1: Dataset statistics.

2 Implementation and Hyperparameters

All methods share the same fixed TIGER teacher checkpoint, RQ-VAE tokenizer, and four-token SID space with codebook size 256. The three student backbones encode the last 20 interactions and score each SID depth with lightweight heads:

- **MLP**: 1.664M trainable parameters;
- **GRU**: hidden size 384, 1.201M trainable parameters;
- **CNN**: embedding dimension 256, 1.753M trainable parameters.

Caveat: the GRU/CNN runs used slightly different parameter budgets than the MLP due to a launch-script issue; a strictly matched rerun is planned and will be reported in the final version.

Students decode with constrained beam search of width $W = 50$ (top-50); the TIGER teacher reference is decoded with beam 20 (the genrec default) and is reported for scale only, since the absolute values are not directly comparable across beam widths.

ServeSID hyperparameters: support size $M = 64$, training cut-off $K_{tr} = 10$, maximum outsider rank $B = 40$, online beam $W = 50$, HServe weight $\lambda_H = 4$, DA-CDRS weight $\lambda_D = 1$, depth weights $\alpha_d = 1$ for $d = 1, \dots, L$, pair margin $\mu = .05$, increment-correction weight $\lambda_{corr} = 31.42$, learning rate 5×10^{-5} , seeds {42, 43, 44}. All reported numbers are three-seed means. Teacher supports, marginals, pairs, and increment scales are built from training requests only; validation/test teacher outputs never enter optimization.

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3 Full Main-Result Tables

Tables 2–7 give the full per-dataset metrics (R@5, N@5, R@10, N@10) and teacher-agreement (Jac@5, Jac@10) for all methods and students.

4 Full Ablation Tables

Tables 8–10 give all six metrics for every ablation arm on each dataset. Ablation arms are defined in the main paper (Section 6.3).

5 Reproducibility Notes

- Evaluation: next-item utility is Recall@ K and NDCG@ K under leave-one-out on the held-out interaction; teacher-agreement Jac@ K is the Jaccard between the student’s beam-50 top- K list and the teacher’s beam-20 top- K list.
- The base student (SID-MLP / base KD) minimizes ground-truth SID cross-entropy plus gold-prefix token KL (Eq. 2 of the main paper). ServeSID replaces the token KL with HServe and DA-CDRS.
- Held-out diagnostic pairs (teacher ranks $1:K$ vs. $K+1:B$) are used only for mechanism statistics and never enter optimization.

Method	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
<i>Teacher TIGER reference (beam-20):</i>												
TIGER	0.043	0.028	0.068	0.036	0.043	0.028	0.068	0.036	0.043	0.028	0.068	0.036
SID-MLP (base KD)	0.0429	0.0282	0.0660	0.0356	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
SmartGR	0.0434	0.0286	0.0668	0.0363	0.0426	0.0286	0.0669	0.0361	0.0374	0.0250	0.0594	0.0320
RD	0.0422	0.0281	0.0657	0.0357	0.0433	0.0285	0.0671	0.0362	0.0419	0.0274	0.0649	0.0348
RRD	0.0430	0.0287	0.0659	0.0361	0.0422	0.0281	0.0664	0.0359	0.0420	0.0281	0.0649	0.0354
RCE-KD	0.0434	0.0286	0.0666	0.0363	0.0432	0.0286	0.0671	0.0361	0.0409	0.0276	0.0651	0.0354
LOHRec	0.0414	0.0278	0.0639	0.0350	0.0376	0.0251	0.0576	0.0315	0.0361	0.0242	0.0542	0.0300
GKD	0.0428	0.0286	0.0668	0.0363	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
DistiLLM-2	0.0428	0.0287	0.0666	0.0361	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
SERVE SID	0.0442	0.0292	0.0679	0.0369	0.0441	0.0291	0.0682	0.0368	0.0437	0.0288	0.0686	0.0368

Table 2: Full utility metrics on Beauty for all methods and students.

Method	MLP		GRU		CNN	
	Jac@5	Jac@10	Jac@5	Jac@10	Jac@5	Jac@10
SID-MLP (base KD)	0.0113	0.0154	0.0091	0.0122	0.0075	0.0103
SmartGR	0.0115	0.0155	0.0092	0.0125	0.0075	0.0106
RD	0.0098	0.0141	0.0096	0.0138	0.0092	0.0135
RRD	0.0125	0.0154	0.0116	0.0143	0.0118	0.0145
RCE-KD	0.0112	0.0151	0.0104	0.0140	0.0094	0.0124
LOHRec	0.0087	0.0116	0.0072	0.0088	0.0081	0.0101
GKD	0.0115	0.0157	0.0091	0.0122	0.0075	0.0103
DistiLLM-2	0.0117	0.0158	0.0091	0.0122	0.0075	0.0103
SERVE SID	0.0118	0.0154	0.0116	0.0151	0.0114	0.0150

Table 3: Teacher-agreement (Jaccard with the TIGER beam-20 top-K list) on Beauty for all methods and students.

Method	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
<i>Teacher TIGER reference (beam-20):</i>												
TIGER	0.041	0.025	0.065	0.033	0.041	0.025	0.065	0.033	0.041	0.025	0.065	0.033
SID-MLP (base KD)	0.0374	0.0235	0.0626	0.0315	0.0367	0.0233	0.0565	0.0296	0.0328	0.0210	0.0533	0.0276
SmartGR	0.0381	0.0241	0.0623	0.0319	0.0382	0.0242	0.0604	0.0314	0.0329	0.0206	0.0520	0.0267
RD	0.0380	0.0234	0.0615	0.0309	0.0381	0.0245	0.0614	0.0318	0.0380	0.0241	0.0603	0.0312
RRD	0.0396	0.0245	0.0625	0.0319	0.0381	0.0245	0.0622	0.0322	0.0378	0.0239	0.0585	0.0306
RCE-KD	0.0387	0.0244	0.0631	0.0324	0.0380	0.0246	0.0622	0.0324	0.0364	0.0232	0.0587	0.0303
LOHRec	0.0362	0.0224	0.0596	0.0299	0.0317	0.0201	0.0510	0.0263	0.0320	0.0198	0.0514	0.0260
GKD	0.0373	0.0237	0.0627	0.0318	0.0382	0.0243	0.0613	0.0316	0.0328	0.0210	0.0533	0.0276
DistiLLM-2	0.0375	0.0238	0.0614	0.0315	0.0367	0.0233	0.0565	0.0296	0.0328	0.0210	0.0533	0.0276
SERVE SID	0.0408	0.0254	0.0644	0.0330	0.0388	0.0250	0.0650	0.0334	0.0398	0.0250	0.0639	0.0328

Table 4: Full utility metrics on Toys for all methods and students.

Method	MLP		GRU		CNN	
	Jac@5	Jac@10	Jac@5	Jac@10	Jac@5	Jac@10
SID-MLP (base KD)	0.0062	0.0089	0.0045	0.0066	0.0038	0.0056
SmartGR	0.0065	0.0089	0.0045	0.0068	0.0041	0.0061
RD	0.0054	0.0080	0.0051	0.0077	0.0047	0.0076
RRD	0.0061	0.0084	0.0057	0.0076	0.0056	0.0080
RCE-KD	0.0061	0.0087	0.0052	0.0075	0.0043	0.0066
LOHRec	0.0054	0.0072	0.0035	0.0048	0.0035	0.0051
GKD	0.0065	0.0091	0.0048	0.0072	0.0038	0.0056
DistiLLM-2	0.0066	0.0092	0.0045	0.0066	0.0038	0.0056
SERVE SID	0.0062	0.0087	0.0060	0.0086	0.0059	0.0083

Table 5: Teacher-agreement (Jaccard with the TIGER beam-20 top-K list) on Toys for all methods and students.

Method	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
<i>Teacher TIGER reference (beam-20):</i>												
TIGER	0.044	0.028	0.070	0.036	0.044	0.028	0.070	0.036	0.044	0.028	0.070	0.036
SID-MLP (base KD)	0.0588	0.0383	0.0922	0.0488	0.0591	0.0389	0.0935	0.0499	0.0553	0.0358	0.0870	0.0460
SmartGR	0.0593	0.0388	0.0929	0.0496	0.0594	0.0391	0.0933	0.0500	0.0549	0.0359	0.0870	0.0462
RD	0.0575	0.0372	0.0927	0.0485	0.0595	0.0388	0.0938	0.0498	0.0580	0.0373	0.0921	0.0483
RRD	0.0592	0.0390	0.0918	0.0496	0.0596	0.0393	0.0892	0.0490	0.0591	0.0389	0.0884	0.0483
RCE-KD	0.0593	0.0388	0.0926	0.0497	0.0595	0.0394	0.0938	0.0503	0.0589	0.0387	0.0919	0.0493
LOHRec	0.0584	0.0377	0.0908	0.0481	0.0538	0.0346	0.0818	0.0436	0.0464	0.0297	0.0711	0.0376
GKD	0.0593	0.0388	0.0929	0.0496	0.0597	0.0393	0.0937	0.0502	0.0569	0.0370	0.0896	0.0475
DistiLLM-2	0.0594	0.0387	0.0927	0.0494	0.0596	0.0392	0.0939	0.0503	0.0587	0.0384	0.0914	0.0489
SERVEID	0.0604	0.0396	0.0944	0.0505	0.0607	0.0400	0.0954	0.0512	0.0605	0.0399	0.0944	0.0508

Table 6: Full utility metrics on VG for all methods and students.

Method	MLP		GRU		CNN	
	Jac@5	Jac@10	Jac@5	Jac@10	Jac@5	Jac@10
SID-MLP (base KD)	0.0100	0.0131	0.0088	0.0117	0.0092	0.0121
SmartGR	0.0100	0.0130	0.0089	0.0118	0.0091	0.0121
RD	0.0086	0.0116	0.0083	0.0113	0.0081	0.0112
RRD	0.0094	0.0114	0.0090	0.0105	0.0090	0.0109
RCE-KD	0.0097	0.0126	0.0091	0.0119	0.0092	0.0120
LOHRec	0.0096	0.0127	0.0079	0.0101	0.0069	0.0091
GKD	0.0101	0.0131	0.0089	0.0118	0.0093	0.0124
DistiLLM-2	0.0100	0.0131	0.0093	0.0121	0.0097	0.0127
SERVEID	0.0094	0.0122	0.0092	0.0121	0.0092	0.0121

Table 7: Teacher-agreement (Jaccard with the TIGER beam-20 top-K list) on VG for all methods and students.

Arm	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
SID-MLP (base KD)	0.0429	0.0282	0.0660	0.0356	0.0416	0.0281	0.0656	0.0358	0.0380	0.0254	0.0580	0.0319
d4-only KL	0.0431	0.0283	0.0663	0.0359	0.0422	0.0283	0.0660	0.0360	0.0402	0.0266	0.0625	0.0348
DA-CDRS only	0.0435	0.0286	0.0672	0.0362	0.0430	0.0286	0.0673	0.0364	0.0414	0.0273	0.0652	0.0359
HServe only	0.0436	0.0287	0.0670	0.0363	0.0431	0.0286	0.0671	0.0364	0.0418	0.0276	0.0656	0.0360
HServe + Uniform	0.0437	0.0286	0.0673	0.0364	0.0437	0.0287	0.0674	0.0365	0.0423	0.0280	0.0669	0.0362
SERVEID(full)	0.0442	0.0292	0.0679	0.0369	0.0441	0.0291	0.0682	0.0368	0.0437	0.0288	0.0686	0.0368

Table 8: Full ablation metrics on Beauty for all students.

Arm	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
SID-MLP (base KD)	0.0374	0.0235	0.0626	0.0315	0.0367	0.0233	0.0565	0.0296	0.0328	0.0210	0.0533	0.0276
d4-only KL	0.0379	0.0237	0.0628	0.0317	0.0369	0.0237	0.0594	0.0313	0.0354	0.0225	0.0566	0.0305
DA-CDRS only	0.0389	0.0246	0.0638	0.0324	0.0377	0.0241	0.0622	0.0323	0.0370	0.0236	0.0598	0.0322
HServe only	0.0390	0.0245	0.0633	0.0323	0.0379	0.0243	0.0630	0.0325	0.0375	0.0238	0.0606	0.0320
HServe + Uniform	0.0395	0.0248	0.0640	0.0325	0.0381	0.0247	0.0642	0.0328	0.0382	0.0242	0.0630	0.0323
SERVEID(full)	0.0408	0.0254	0.0644	0.0330	0.0388	0.0250	0.0650	0.0334	0.0398	0.0250	0.0639	0.0328

Table 9: Full ablation metrics on Toys for all students.

Arm	MLP				GRU				CNN			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10
SID-MLP (base KD)	0.0588	0.0383	0.0922	0.0488	0.0591	0.0389	0.0935	0.0499	0.0553	0.0358	0.0870	0.0460
d4-only KL	0.0589	0.0384	0.0925	0.0491	0.0593	0.0390	0.0937	0.0502	0.0570	0.0370	0.0888	0.0488
DA-CDRS only	0.0595	0.0388	0.0932	0.0498	0.0601	0.0394	0.0943	0.0507	0.0586	0.0380	0.0914	0.0498
HServe only	0.0597	0.0390	0.0935	0.0499	0.0600	0.0395	0.0945	0.0506	0.0591	0.0386	0.0923	0.0500
HServe + Uniform	0.0599	0.0391	0.0938	0.0500	0.0602	0.0397	0.0950	0.0510	0.0596	0.0390	0.0935	0.0502
SERVEID(full)	0.0604	0.0396	0.0944	0.0505	0.0607	0.0400	0.0954	0.0512	0.0605	0.0399	0.0944	0.0508

Table 10: Full ablation metrics on VG for all students.